Evaluating Adaptive Region Acquisition for Spatial Transcriptomics Reconstruction

Ayush Deo
University of Southern California

Abstract

Region-based spatial transcriptomics measures only part of a tissue section, and several recent methods choose each next region from the expression already measured. We asked whether this feedback helps, for two objectives that reward different choices: reconstructing expression across the section and capturing regions of high molecular heterogeneity. In a retrospective simulator built from 22 human HEST-1k slides (18 studies, four tissues), we held the image encoder and imputer fixed and compared value-conditioned policies with a non-adaptive placement rule chosen on held-out studies. The analyses were registered before each experimental stage and resample whole studies. For reconstruction, a value-conditioned policy designed to target the imputer's prediction error did worse than the non-adaptive rule at every budget from 0.5% to 20% of regions (mean differences -0.092 to -0.002 PCC), although feedback improved it over the same policy given permuted values at the smallest budgets. A semi-synthetic control with injected hotspot genes also showed no gain, so the protocol's sensitivity to reconstruction gains remains unestablished. For detection, informative feedback raised recall over permuted feedback by $+0.005$ to $+0.063$, but no difference survived correction for multiple comparisons, the larger estimates fell below the minimum detectable effect, and a non-adaptive diversity sampler performed about as well. Within this retrospective setting, with computationally defined regions of interest and spot-sized acquisition units, the tested value-conditioned policies did not improve reconstruction, and whether feedback improves detection remains unresolved.

1 Introduction

Whole-transcriptome spatial assays are expensive, and region-of-interest platforms such as GeoMx DSP [9] profile only part of a section. The rest can be predicted afterwards, usually from co-registered hematoxylin and eosin (H&E) images [5, 11]. Which regions are measured therefore shapes two outcomes: how well the unmeasured tissue can be reconstructed, and whether biologically distinct regions are measured at all.

Recent methods choose regions sequentially. SOFisher [7] trains a reinforcement-learning policy that moves fields of view toward regions enriched for a target signal, using the expression measured so far. SCR^2-ST [15] also uses reinforcement learning, choosing sampling locations for expression prediction with a reward dominated by coverage of a single-cell reference. S2-omics [14] instead selects regions from histology before any measurement, aiming to capture molecular heterogeneity, and its preprint argues that waiting for measured expression between choices is impractical. These methods serve different goals. SOFisher and S2-omics try to capture particular regions, while reconstruction tries to predict expression everywhere.

The two goals lead to different expectations about feedback. For reconstruction, Gaussian-process sensor placement suggests that feedback should matter little: with a fixed kernel, posterior variance

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depends on where measurements are taken and not on the values observed, so the best adaptive and non-adaptive designs coincide [6]. Detection has no comparable argument, because the targets are defined by the values themselves. Evaluating either goal also requires a fair baseline. A placement rule that could be chosen before profiling a new slide is the relevant competitor, not the best rule for that slide picked in hindsight.

We therefore asked three questions with the image encoder and imputer held fixed: **Q1**, does conditioning region choice on measured expression improve whole-section reconstruction over a feasible non-adaptive rule? **Q2**, does informative feedback improve the capture of molecularly heterogeneous regions? **Q3**, does any detection advantage survive the batched and contiguous acquisition that real platforms impose? To separate the contribution of feedback from that of a selection rule, we compared each value-conditioned policy with the same policy given uninformative values. To learn whether a null result could have been otherwise, we ran synthetic and semi-synthetic positive controls.

None of the tested value-conditioned policies improved reconstruction, and the semi-synthetic control did not establish that the protocol would detect a reconstruction gain if one existed. Feedback produced positive but uncertain detection gains that a non-adaptive diversity design matched. The study contributes an evaluation rather than a method: a reproducible protocol on 22 human slides from 18 studies, analyses registered before each experimental stage, and the finding that the tested feedback policies offered no supported advantage over feasible non-adaptive placement for either objective.

2 Related work

Expression has been predicted from H&E with contrastive retrieval [11], flow matching over spatial neighbourhoods [4], diffusion [16], and autoregressive generation [10], and benchmarked on HEST-1k [5]. We hold the predictor fixed and vary only the regions it receives. For deciding what to measure, gene-panel selectors fix a set of genes [2, 8], and region selectors range from one-shot image rankers [12] and histology-based heterogeneity selection [14] to sequential reinforcement-learning samplers [7, 15]. Active feature acquisition casts measurement as a sequential policy [13]. Variance- and mutual-information-greedy placement are near-optimal for reconstructing a Gaussian-process field [6] and serve here as non-adaptive baselines.

3 Methods

3.1 Slides and acquisition unit

We used 22 human HEST-1k slides [5]: prostate (6), kidney (4), lung (6), and colon (6), of which 7 are Visium HD and 15 standard Visium (Appendix A). Selection rules were fixed before any result on these slides: human tissue, whole-transcriptome assay, at least 1,000 spots under tissue, a source image of at most 0.6 μm/px so that image inputs are comparable, one slide per specimen, and at most two slides per study to limit dependence. The slides come from 18 HEST studies, which we treat as the independent units.

A region is one native spot. Visium spots are 55 μm wide on a 100 μm pitch; the Visium HD slides, as distributed by HEST, have a median spacing of 128 μm on our slides. We used the spot because HEST provides expression and an image patch at that unit, and because a fixed square grid would pool different numbers of spots on slides with different pixel sizes. Slides contain 1,766 to 9,080 regions. The image input is the HEST 224 px patch at 0.5 μm/px centred on each spot, embedded with frozen UNI2-h [1]. Expression is log1p-transformed, and genes detected in fewer than 10% of regions are removed.

3.2 Objectives

A policy reveals regions up to a budget, given as a fraction of the slide's regions (0.5% to 20%), and each revealed region exposes its measured expression. For *reconstruction* (Q1), a fixed imputer predicts the unprofiled regions and we score the per-gene Pearson correlation over those regions, averaged over the 50 most variable genes (PCC), the HEST-Benchmark target [5]. For *detection* (Q2,

Q3), regions of interest (ROIs) are the 20% of regions with the highest local molecular heterogeneity, the variance of top-50-gene expression across a region's 8 nearest spatial neighbours, following the heterogeneity objective of S2-omics [14]. We score recall, the fraction of ROIs that were profiled. These ROIs are computational; they are not annotated regions of biological interest. Gene sets and ROIs come from ground truth and are used only for scoring.

3.3 Imputer

The imputer (graphflow) maps image principal components through 512 random Fourier features, fits a ridge regression to the profiled regions, and runs 20 steps of message passing on a fixed 8-nearest-neighbour graph in image and spatial features, holding profiled regions at their measured values. One fixed imputer serves every policy, so differences between policies reflect selection alone. Its predictions depend on the measured values, but its uncertainty, one minus the graph influence that reaches a region from profiled nodes, depends only on which regions have been profiled. A k-nearest-neighbour and a regression-kriging imputer were run on an earlier slide set (Appendix H).

3.4 Policies

Non-adaptive rules: random, farthest-point in space (grid), k-means on image embeddings with medoids, a reimplementation of a one-shot image ranker [12], and variance-greedy and mutual-information (MI)-greedy placement with an RBF kernel on image and coordinate features [6]. *Sequential fixed designs*: error-reduction picks the region whose 15 nearest unprofiled neighbours (in the imputer's feature space) have the largest summed uncertainty, and maximum-uncertainty picks the most uncertain region. Under graphflow neither depends on measured values (for error-reduction we confirmed on a test slide that permuted values yield identical selections), so they are fixed designs computed one region at a time. *Value-conditioned policies*: the *residual* policy keeps the error-reduction form but replaces the uncertainty with an estimate of the imputer's actual error. It takes the leave-one-out residual of the image ridge model at each profiled region, averages it over each unprofiled region's 10 nearest profiled neighbours, and multiplies by one minus the graph influence, so it targets reconstruction error through the measured values. The *local-variance* policy scores a candidate by the kernel-weighted variance of measured expression around it plus 0.5 times its distance to the nearest profiled region; it seeks heterogeneous areas and is the detection policy. *Competitor objectives*: a non-adaptive diversity sampler (farthest-point in image and space, after the coverage-dominated reward of SCR^2-ST [15]) and an adaptive ROI-seeking sampler after SOFisher [7]. These reimplement stated objectives and are not the authors' trained models.

To isolate feedback, each value-conditioned policy was also run with *permuted* feedback, in which it sees the expression of a randomly chosen other region whenever it profiles a region, and with *constant* feedback, in which every region appears to have the slide's mean expression. The selection rule is unchanged; only the information in the values is removed. The imputer and the scores always use the true values.

3.5 Comparators

The primary comparator is a *feasible* rule chosen without the target slide. For each held-out study, we select the non-adaptive rule (grid, k-means, variance-greedy, MI-greedy, or one-shot) with the best mean score on the other 17 studies at that budget and objective, and apply it to the held-out slides. It selected k-means for reconstruction at 0.5–1% and MI-greedy at 2–20% (with k-means for 4 of 18 studies at 2%), and for detection variance-greedy for most studies at 2–10% and the one-shot ranker at 20% (Appendix C). The per-slide best rule, which uses each slide's outcome, is kept only as a hindsight benchmark. For detection under batching (Q3) the comparator is the per-slide best of k-means, grid, and MI-greedy, a hindsight comparator that favours the baselines.

3.6 Statistical analysis

Each margin is a per-slide difference between two policies. We report its mean, a 95% percentile bootstrap CI that resamples whole studies (5,000 resamples), and an exact two-sided sign-flip test that flips whole studies (all 2^{18} sign patterns). The CIs are not adjusted for multiplicity; p_H denotes Holm-adjusted p-values [3]. We also report the minimum detectable effect (MDE): the smallest

true mean margin that the test would detect with 80% power at $\alpha = 0.05$, estimated by resampling studies around the observed dispersion.

The analyses were registered in the repository in two stages, each before the corresponding experiments were run. Stage 1 compared policies with per-slide best rules in a family of 24 margins (Appendix B); the batched and contiguous experiments for Q3 come from this stage. Stage 2 was registered after the stage-1 results were known and before its experiments. It introduced the feasible comparator, the residual policy, and the feedback ablations, with a primary family of 18 margins: residual minus feasible for reconstruction at six budgets (Q1), and for detection at four budgets, local-variance minus its permuted-feedback version, minus the feasible rule, and minus the diversity sampler (Q2). The decision rules were to claim a benefit only for a positive margin with $p_H < 0.05$ and to report any other outcome as not established. Results are labelled confirmatory (registered primary), secondary (registered, Holm within the secondary family), descriptive, or post hoc.

3.7 Positive controls

A null result is informative only if the protocol could have registered an effect. We used two controls. In synthetic slides of 20×20 regions with image-unpredictable, spatially clustered structure (8 slides per setting, 10% budget, k-NN imputer), we varied the correlation length of the hidden structure and the amplitude α with which ROIs appear in measured expression (descriptive). On the 22 real slides we added 20 genes carrying three hotspots (radius 5% of the slide diagonal) that the image cannot predict, scaled so that each gene's standard deviation is α times the median of the slide's top-50 genes, with $\alpha \in \{0, 1, 2\}$. Before running it, we set the criterion that the control succeeds if the residual policy beats the feasible rule with a CI excluding zero at $\alpha = 2$.

4 Results

4.1 Positive controls

The synthetic controls showed that the simulator can register an effect of feedback on detection. The local-variance policy's recall margin over coverage rose with ROI amplitude, from -0.045 at $\alpha = 0$ to $+0.052$ (95% CI $+0.033$ to $+0.071$) at $\alpha = 4$ and $+0.101$ ($+0.065$ to $+0.135$) at $\alpha = 8$, positive on all 8 slides at both amplitudes (Figure 1b). Reconstruction behaved differently. A non-deployable ordering by true local difficulty stayed below coverage at every correlation length, by 0.105 and 0.066 PCC at the two shortest lengths and 0.013 to 0.020 at the longer ones (Figure 1a).

The semi-synthetic control on real slides did not meet its criterion. At $\alpha = 1$ and 2 all 20 injected genes entered the evaluation set on every slide, yet the residual policy remained below the feasible rule: -0.021 (-0.030 to -0.012) at 5% and -0.011 (-0.019 to -0.004) at 10% for $\alpha = 2$, close to its deficits without injection (-0.015 and -0.007; Figure 2c, Appendix E). Feedback still helped relative to permuted values, by $+0.001$ to $+0.013$. The protocol can therefore register a large detection effect, but we could not show that it would register a reconstruction gain, and the Q1 results below should be read as statements about the tested policies.

4.2 Q1: Reconstruction

No value-conditioned policy outperformed the feasible rule (Tables 1 and 2, Figure 2a). The residual policy was below it at every budget, by 0.092 PCC at 0.5%, 0.057 at 1%, 0.025 at 2%, 0.016 at 5%, 0.007 at 10%, and 0.002 at 20%. The deficit was significant at 0.5, 1, 5, and 10% ($p_H \leq 0.010$). Under the registered rule, Q1 has no benefit to report. The local-variance policy, which seeks heterogeneous areas, was further below, by 0.119 to 0.163 PCC (secondary, $p_H \leq 0.001$).

The feedback ablation shows that measured values did carry information for reconstruction. Given informative rather than permuted values, the residual policy improved by $+0.031$, $+0.039$, and $+0.030$ PCC at 0.5–2% (secondary; p_H 0.056, 0.002, and 0.014), with no difference at 10–20%. Feedback thus moved the residual policy toward the feasible rule without closing the gap. The policy's error-reduction structure is the likely reason. The sequential fixed design with the same structure but value-independent uncertainty trailed the feasible rule by similar amounts at low budgets (-0.107, -0.067, and -0.024 at 0.5–2%; post hoc), and both converged to it as the budget grew.

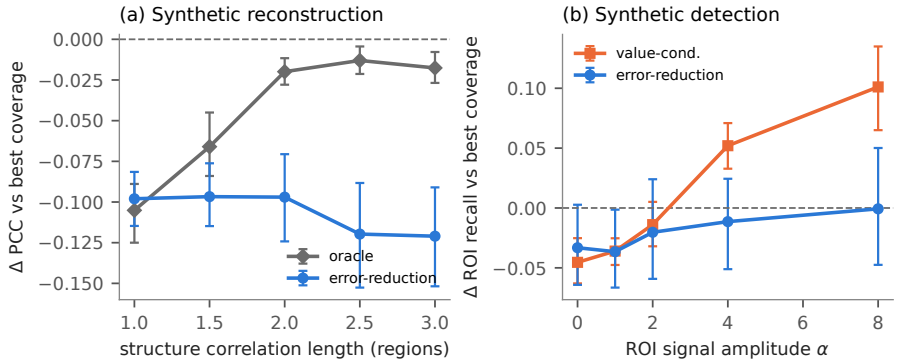


Figure 1: Synthetic controls (8 slides per point, 10% budget, mean and 95% bootstrap CI over slides). (a) Reconstruction margin over the best coverage rule versus the correlation length of the hidden structure, for an ordering by true local difficulty and for error-reduction. (b) Detection-recall margin versus ROI amplitude.

Table 1: Reconstruction accuracy (PCC over the top-50 genes on unprofiled regions), mean over 22 slides. "Feasible" is the non-adaptive rule chosen on the other studies; "per-slide best" uses each slide's outcome. "Residual, perm." is the residual policy given permuted feedback.

	Non-adaptive		Sequential fixed design		Value-conditioned		
Budget	feasible	per-slide best	error-red.	max-unc.	residual	residual, perm.	local-var.
0.5%	0.449	0.470	0.343	0.348	0.357	0.327	0.348
1%	0.534	0.550	0.467	0.462	0.477	0.438	0.424
2%	0.591	0.607	0.567	0.562	0.566	0.536	0.471
5%	0.659	0.660	0.650	0.648	0.643	0.633	0.513
10%	0.685	0.687	0.684	0.685	0.678	0.677	0.535
20%	0.707	0.709	0.708	0.710	0.705	0.705	0.544

At 20% the maximum-uncertainty design was $+0.003$ PCC above the feasible rule (95% CI $+0.002$ to $+0.004$, $p = 0.0007$, post hoc), a small advantage of a fixed design that involves no feedback.

The hindsight benchmark changed the size of these deficits but not their direction. The per-slide best rule exceeded the feasible rule by 0.016 to 0.021 PCC at budgets of 2% or less and by at most 0.002 above that (Appendix C).

4.3 Q2: Detection

Informative feedback raised the local-variance policy's recall over permuted feedback by $+0.005$, $+0.017$, $+0.029$, and $+0.063$ at 2, 5, 10, and 20% (Table 3, Figure 2b). The unadjusted CIs exclude zero at 5–20%, but none survived adjustment ($p_H \geq 0.29$). These estimates lie below the minimum detectable effects of 0.026, 0.041, and 0.084 at those budgets, so the experiment could miss effects of this size. The margin over the feasible rule was smaller ($+0.006$ to $+0.026$, $p_H \geq 0.25$), and the margin over the diversity sampler was close to zero ($+0.001$ to $+0.012$, $p_H = 1.000$).

The absolute recalls show why the feedback margin exceeds the margin over the feasible rule (Table 4). Without informative feedback the local-variance policy fell below the feasible rule (0.215 with permuted and 0.204 with constant values, against 0.252 at 20%), and informative feedback lifted it to 0.278. Part of the feedback effect therefore recovers ground the selection rule loses, rather than adding to what a good non-adaptive design achieves. The diversity sampler reached 0.266 without any feedback. Against the exploration-only variant, feedback was significant at 20% ($+0.074$, $p_H = 0.038$, secondary). The ROI-seeking sampler did not improve recall over coverage in stage 1 (Appendix B). Because that port was not the trained SOFisher, we also ran SOFisher natively on its own mouse cortex data. Its evaluation reward for captured target regions rose from a median of -238 early in training to $+147$ late (descriptive; Appendix I). The port's result on our

Table 2: Reconstruction margins (PCC), 22 slides, 18 study clusters. Δ is the mean per-slide difference with an unadjusted 95% CI that resamples studies. p_H: Holm-adjusted within the stage-2 primary or secondary family; p: unadjusted (post hoc).

	residual − feasible (primary)		residual − residual, perm. (secondary)		error-red. − feasible (post hoc)	
Budget	Δ [95% CI]	p_H	Δ [95% CI]	p_H	Δ [95% CI]	p
0.5%	-0.092 $[-0.132, -0.054]$	0.002	$+0.031$ $[+0.010, +0.057]$	0.056	-0.107 $[-0.149, -0.067]$	<0.001
1%	-0.057 $[-0.089, -0.028]$	0.010	$+0.039$ $[+0.021, +0.060]$	0.002	-0.067 $[-0.093, -0.044]$	<0.001
2%	-0.025 $[-0.041, -0.009]$	0.099	$+0.030$ $[+0.014, +0.049]$	0.014	-0.024 $[-0.039, -0.007]$	0.013
5%	-0.016 $[-0.025, -0.009]$	0.003	$+0.009$ $[+0.001, +0.018]$	0.238	-0.010 $[-0.017, -0.003]$	0.021
10%	-0.007 $[-0.011, -0.003]$	0.010	$+0.001$ $[-0.002, +0.004]$	1.000	-0.001 $[-0.006, +0.003]$	0.763
20%	-0.002 $[-0.004, -0.001]$	0.126	-0.001 $[-0.003, +0.002]$	1.000	$+0.001$ $[-0.002, +0.003]$	0.537

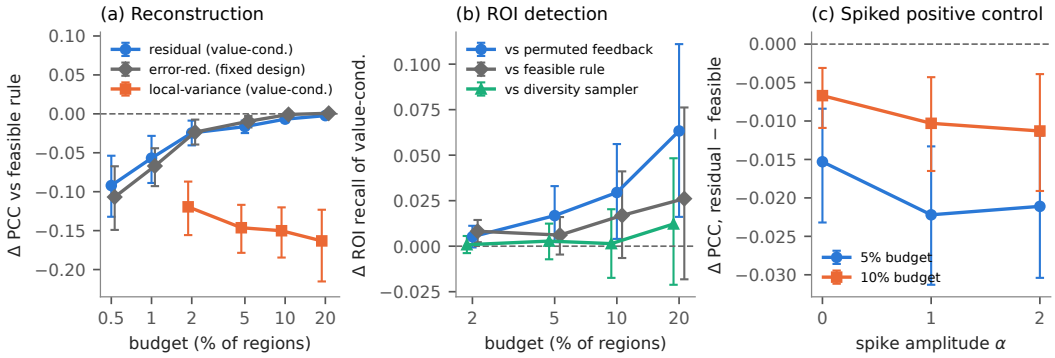


Figure 2: Margins over 22 slides, mean and unadjusted 95% CI resampling studies. (a) Reconstruction margin over the feasible rule. (b) Detection-recall margin of the local-variance policy over its permuted-feedback version, the feasible rule, and the diversity sampler. (c) Spiked positive control: residual policy minus feasible rule by spike amplitude.

slides is therefore not a test of SOFisher. SCR^2-ST could not be run natively because its processed training data are not public.

4.4 Q3: Batched and contiguous acquisition

Real platforms do not let a policy see each measurement before choosing the next region. In stage 1 we revealed regions at 10% budget in rounds of 1, 5, or 25, or in one round, with the policy using only measurements available at the start of each round, and separately in contiguous blocks of 7 regions around a chosen seed (Table 5). The local-variance policy's detection margins over the hindsight comparator were $+0.021$, $+0.034$, and $+0.011$ for rounds of 1, 5, and 25 and $+0.024$ for contiguous blocks, none with $p_H < 0.05$ in the stage-1 family. With a single round the policy has no measurements and falls back to its starting rule, which takes the regions nearest the slide centroid. The resulting margin of -0.051 ($p_H = 0.008$) reflects that starting rule more than the value of adaptivity. Reconstruction margins were negative in every setting. Since Q2 found no established advantage, Q3 could not confirm one under realistic acquisition, and the comparator here is the hindsight maximum.

4.5 Robustness

Selecting the evaluation genes from the full slide uses information that a prospective experiment would lack. Selecting them from a random 15% reference subset changed the stage-1 reconstruction margins by at most 0.003, and scoring all filtered genes left them negative (-0.033 to -0.079; Appendix G). In an exploratory split, the stage-1 detection margins were larger on the Visium HD slides than on Visium ($+0.048$ against $+0.008$ at 10%), but with 5 HD studies the smallest attainable p is 0.0625, and platform cannot be separated from study (Appendix F).

Table 3: Detection margins (ROI recall) of the local-variance policy, 22 slides. Δ with unadjusted 95% CI resampling studies; p_H Holm-adjusted over the 18 stage-2 primary margins; MDE is the minimum detectable effect at 80% power, unadjusted.

	value-cond. − permuted feedback			value-cond. − feasible		value-cond. − diversity	
Budget	Δ [95% CI]	p_H	MDE	Δ [95% CI]	p_H	Δ [95% CI]	p_H
2%	$+0.005\ [-0.001, +0.011]$	1.000	0.010	$+0.008\ [+0.002, +0.014]$	0.250	$+0.001\ [-0.004, +0.006]$	1.000
5%	$+0.017\ [+0.002, +0.033]$	0.530	0.026	$+0.006\ [-0.005, +0.016]$	1.000	$+0.003\ [-0.007, +0.012]$	1.000
10%	$+0.029\ [+0.004, +0.056]$	0.530	0.041	$+0.017\ [-0.006, +0.041]$	1.000	$+0.001\ [-0.017, +0.020]$	1.000
20%	$+0.063\ [+0.016, +0.111]$	0.294	0.084	$+0.026\ [-0.018, +0.076]$	1.000	$+0.012\ [-0.021, +0.048]$	1.000

Table 4: ROI-detection recall, mean over 22 slides. "Observed", "permuted", and "constant" are the local-variance policy with informative, permuted, and constant feedback. The oracle profiles regions in order of true heterogeneity and bounds recall from above.

	Non-adaptive				Value-conditioned (local variance)			
Budget	feasible	k-means	MI-greedy	diversity	observed	permuted	constant	oracle
2%	0.018	0.018	0.018	0.026	0.027	0.021	0.022	0.100
5%	0.064	0.050	0.046	0.067	0.070	0.053	0.051	0.250
10%	0.120	0.110	0.094	0.136	0.137	0.107	0.103	0.500
20%	0.252	0.232	0.200	0.266	0.278	0.215	0.204	0.999

5 Discussion

The results support a narrower conclusion than a general verdict on adaptive acquisition. For reconstruction, the tested value-conditioned policies, including one designed to target the imputer's prediction error, did not improve on a non-adaptive rule chosen on other studies at any budget from 0.5% to 20%. The feedback ablation separates two components of this outcome. Measured values carried usable information, since the residual policy improved by up to 0.039 PCC when its feedback was informative. Its error-reduction structure, which held back the value-independent fixed design by similar amounts, left it behind k-means and MI-greedy at low budgets, and the information did not make up the difference. This is consistent with the sensor-placement view that, for reconstruction, where measurements are placed matters more than what they return, but it does not test that view, and the failed semi-synthetic control prevents us from claiming that useful value-conditioned reconstruction is impossible. A policy that combines a strong coverage rule with feedback, rather than one that replaces coverage with feedback, remains untested.

For detection, the question remains open. Feedback improved recall relative to the same policy without informative values, and the estimates grew with budget, but no margin survived correction, the estimates were below the detectable effect sizes, and a non-adaptive diversity design came within 0.012 recall of the adaptive policy. Settling whether feedback adds to a good non-adaptive design would need more independent studies than the 18 used here.

For practice at the scale tested, a non-adaptive rule chosen on other slides was a sufficient baseline. k-means at very low budgets and MI-greedy above them matched or exceeded every value-conditioned policy we tested for reconstruction (only a fixed design, maximum-uncertainty, exceeded MI-greedy, by 0.003 PCC at 20%), and a diversity design was as good as feedback for computationally defined ROIs, without waiting for measurements between rounds. Adaptive acquisition methods should be compared with such baselines on the objective they are meant to serve.

Limitations. The simulator is retrospective: a revealed region returns its measured expression without added noise, batch effects, or turnaround time, conditions that if anything favour feedback. ROIs are defined by a local-variance score, not by annotation, and the acquisition unit is a spot or HD pseudo-spot, which may be smaller than ROIs on platforms such as GeoMx; the HD and Visium units also differ in spacing. The batch and block simulations approximate but do not validate any particular platform. The 22-slide results use one imputer, and the k-NN and kriging comparisons rest on 5 slides. The competitor samplers were reimplementations of objectives; SOFisher was run natively only on its own data and SCR^2-ST not at all. The feasible comparator, the residual policy,

Table 5: Local-variance policy minus the per-slide best coverage rule (hindsight) at 10% budget under batched and contiguous acquisition, stage 1. Δ with unadjusted 95% CI resampling studies; p_H Holm-adjusted over the 24 stage-1 primary margins.

	Reconstruction (PCC)		Detection (ROI recall)	
Acquisition setting	Δ [95% CI]	p_H	Δ [95% CI]	p_H
Sequential (1 region per round)	-0.152 $[-0.186, -0.122]$	<0.001	$+0.021$ $[+0.003, +0.037]$	0.362
Rounds of 5 regions	-0.129 $[-0.175, -0.094]$	<0.001	$+0.034$ $[+0.013, +0.053]$	0.073
Rounds of 25 regions	-0.130 $[-0.176, -0.095]$	<0.001	$+0.011$ $[-0.000, +0.021]$	0.362
Single round (whole budget)	-0.236 $[-0.288, -0.188]$	<0.001	-0.051 $[-0.077, -0.027]$	0.008
Contiguous blocks of 7 regions	-0.087 $[-0.116, -0.063]$	<0.001	$+0.024$ $[+0.002, +0.044]$	0.362

and the feedback ablations were added after the stage-1 results were known, although they were registered before their experiments. Budgets above 20% were not tested.

6 Conclusion

Q1: across 22 human slides from 18 studies, the tested value-conditioned policies did not improve whole-section reconstruction over a feasible non-adaptive rule at any budget from 0.5% to 20%. Feedback improved an error-targeting policy relative to uninformative values at low budgets, but not enough to match the non-adaptive rule, and a semi-synthetic control did not establish that the protocol could detect a reconstruction gain. Q2: whether feedback improves the detection of computationally defined heterogeneous regions remains unresolved. Estimated gains of up to 0.063 recall did not survive correction and fell below the detectable effect size, and a non-adaptive diversity sampler performed comparably. Q3: no batched or contiguous setting yielded an established detection advantage. Within this retrospective, spot-level setting, the tested feedback policies offered no supported advantage over feasible non-adaptive placement.

Reproducibility. Code, configuration, result files, both registered analysis plans, and the scripts that regenerate every table and figure are available at https://github.com/ayushdeo/STARE.

A Slides

Table 6 lists the slides. "Cluster" indexes the HEST study used as the resampling unit, and "Pitch" is the median nearest-neighbour spacing of regions.

Table 6: The 22 slides (HEST-1k v1.3 metadata).

Tissue	HEST ID	Platform	Spots	μm/px	Pitch (μm)	Cluster	Study (HEST dataset title)
prostate	TENX174	Visium HD	2385	0.274	128	1	Visium HD Spatial Gene Expression Library, Human Prostate ...
prostate	TENX40	Visium	4371	0.345	100	2	Human Prostate Cancer, Adenocarcinoma with Invasive Carcin...
prostate	MEND160	Visium	4079	0.573	100	3	Spatially resolved clonal copy number alterations in benig...
prostate	TENX46	Visium	3043	0.344	100	4	Human Prostate Cancer, Acinar Cell Carcinoma (FFPE)
prostate	TENX41	Visium	2543	0.345	100	5	Normal Human Prostate (FFPE)
prostate	MEND144	Visium	4074	0.341	100	3	Spatially resolved clonal copy number alterations in benig...
kidney	TENX173	Visium HD	2684	0.274	128	6	Visium HD Spatial Gene Expression Library, Human Kidney (F...
kidney	TENX71	Visium	5936	0.274	100	7	Human Kidney, 11 mm Capture Area (FFPE)
kidney	INT21	Visium	4975	0.457	100	8	Tertiary lymphoid structures generate and propagate anti-t...
kidney	INT19	Visium	4948	0.457	100	8	Tertiary lymphoid structures generate and propagate anti-t...
lung	TENX171	Visium HD	1766	0.274	128	9	Visium HD Spatial Gene Expression Libraries, Post-Xenium, ...
lung	TENX170	Visium HD	1877	0.274	128	9	Visium HD Spatial Gene Expression Libraries, Post-Xenium, ...
lung	TENX163	Visium HD	2487	0.274	128	10	Visium HD Spatial Gene Expression Library, Human Lung Canc...
lung	TENX72	Visium	6195	0.274	100	11	Human Lung Cancer, 11 mm Capture Area (FFPE)
lung	TENX62	Visium	3858	0.273	100	12	Human Lung Cancer (FFPE)
lung	MISC20	Visium	3316	0.438	100	13	A spatially resolved atlas of the human lung characterizes...
colon	TENX156	Visium HD	2179	0.274	128	14	Characterization of immune cell populations in the tumor m...
colon	TENX155	Visium HD	2281	0.274	128	14	Characterization of immune cell populations in the tumor m...
colon	TENX70	Visium	9080	0.274	100	15	Human Colorectal Cancer, 11 mm Capture Area (FFPE)
colon	TENX92	Visium	6518	0.274	100	16	Visium CytAssist Gene Expression Libraries of Post-Xenium ...
colon	MISC47	Visium	4818	0.276	100	17	COLON MAP: Colon Molecular Atlas Project
colon	ZEN48	Visium	2385	0.459	100	18	Charting the Heterogeneity of Colorectal Cancer Consensus ...

Changes from an earlier seven-slide analysis. An earlier analysis used seven Visium HD slides and described regions as 112 μm squares. The current HEST metadata lists two of those slides as

non-human (TENX186, *Mus musculus*; TENX175, mixed human and mouse), and both are excluded here. The loader had also assumed 0.5 μm/px, whereas the HD images are about 0.274 μm/px, so those regions were about 61 μm squares containing one pseudo-spot each. On the five retained slides, per-slide local-variance margins (reconstruction and detection, four budgets) differ between the two analyses by a median of 0.015 (maximum 0.096); 3 of 40 change sign, two by less than 0.005 and one detection margin from -0.050 to $+0.015$. With seven slides the smallest attainable exact p-value is 0.016, so no earlier margin could have survived correction for 24 comparisons.

B Stage-1 confirmatory results

Stage 1 compared policies with the per-slide best rule (hindsight). Table 7 gives its 24-margin primary family except the batched settings shown in Table 5; Table 8 gives absolute recall, including the ROI-seeking sampler.

Table 7: Stage-1 primary margins (hindsight comparators). Error-reduction is compared with the per-slide best of five non-adaptive rules, local-variance with the per-slide best of k-means, grid, and MI-greedy. Unadjusted 95% CIs resampling studies; p_H Holm-adjusted over the 24 stage-1 primary margins.

	Reconstruction (PCC, top-50 genes)				Detection (ROI recall)	
	error-reduction − best non-adapt.		value-cond. − best coverage		value-cond. − best coverage	
Budget	Δ [95% CI]	p_H	Δ [95% CI]	p_H	Δ [95% CI]	p_H
0.5%	-0.128 $[-0.162, -0.096]$	<0.001	–	–	–	–
1%	-0.083 $[-0.105, -0.064]$	<0.001	–	–	–	–
2%	-0.040 $[-0.054, -0.027]$	<0.001	-0.136 $[-0.171, -0.106]$	<0.001	$+0.001$ $[-0.004, +0.006]$	0.723
5%	-0.011 $[-0.018, -0.004]$	0.057	-0.148 $[-0.180, -0.119]$	<0.001	$+0.013$ $[+0.002, +0.024]$	0.362
10%	-0.002 $[-0.007, +0.001]$	0.687	-0.152 $[-0.186, -0.122]$	<0.001	$+0.021$ $[+0.003, +0.037]$	0.362
20%	-0.002 $[-0.004, +0.001]$	0.687	-0.165 $[-0.217, -0.125]$	<0.001	$+0.042$ $[+0.006, +0.077]$	0.362

Table 8: Stage-1 ROI-detection recall, mean over 22 slides. †Reimplementations of the stated objectives of SCR^2-ST (diversity, non-adaptive) and SOFisher (ROI-seeking, adaptive). *Ground-truth ordering.

	Non-adaptive coverage			Competitor objectives			
Budget	k-means	grid	MI-greedy	diversity†	ROI-seeking†	value-cond.	oracle*
2%	0.018	0.022	0.018	0.026	0.021	0.027	0.100
5%	0.050	0.051	0.046	0.067	0.050	0.070	0.250
10%	0.110	0.103	0.094	0.136	0.104	0.137	0.500
20%	0.232	0.204	0.200	0.266	0.232	0.278	0.999

C Feasible comparator choices and the hindsight gap

Table 9: Rules chosen on the other studies (number of held-out studies, of 18) and the difference between the per-slide best rule and the feasible rule for reconstruction (unadjusted 95% CI resampling studies).

Budget	feasible choice, reconstruction	feasible choice, detection	per-slide best − feasible (PCC)
0.5%	kmeans (18)	–	$+0.021$ $[+0.006, +0.038]$
1%	kmeans (18)	–	$+0.016$ $[+0.006, +0.027]$
2%	mi-greedy (14), kmeans (4)	var-greedy (12), oneshot (5), grid (1)	$+0.017$ $[+0.008, +0.027]$
5%	mi-greedy (18)	var-greedy (18)	$+0.001$ $[+0.000, +0.003]$
10%	mi-greedy (18)	var-greedy (16), oneshot (2)	$+0.001$ $[+0.001, +0.003]$
20%	mi-greedy (18)	oneshot (18)	$+0.002$ $[+0.001, +0.003]$

D Stage-2 secondary family

Table 10: Stage-2 secondary margins, 22 slides. Unadjusted 95% CIs resampling studies; p_H Holm-adjusted over the secondary family.

Margin	Budget	Δ [95% CI]	p_H
residual − residual, perm. (PCC)	0.5%	$+0.031$ $[+0.010, +0.057]$	0.056
residual − per-slide best (PCC)	0.5%	-0.113 $[-0.146, -0.084]$	<0.001
residual − residual, perm. (PCC)	1%	$+0.039$ $[+0.021, +0.060]$	0.002
residual − per-slide best (PCC)	1%	-0.072 $[-0.099, -0.051]$	<0.001
residual − residual, perm. (PCC)	2%	$+0.030$ $[+0.014, +0.049]$	0.014
residual − per-slide best (PCC)	2%	-0.041 $[-0.056, -0.029]$	<0.001
value-cond. − value-cond., constant (recall)	2%	$+0.004$ $[-0.001, +0.010]$	0.518
value-cond. − feasible (PCC)	2%	-0.119 $[-0.156, -0.087]$	<0.001
residual − residual, perm. (PCC)	5%	$+0.009$ $[+0.001, +0.018]$	0.238
residual − per-slide best (PCC)	5%	-0.018 $[-0.026, -0.011]$	<0.001
value-cond. − value-cond., constant (recall)	5%	$+0.019$ $[+0.006, +0.032]$	0.083
value-cond. − feasible (PCC)	5%	-0.146 $[-0.178, -0.117]$	<0.001
residual − residual, perm. (PCC)	10%	$+0.001$ $[-0.002, +0.004]$	1.000
residual − per-slide best (PCC)	10%	-0.008 $[-0.012, -0.005]$	<0.001
value-cond. − value-cond., constant (recall)	10%	$+0.034$ $[+0.011, +0.057]$	0.076
value-cond. − feasible (PCC)	10%	-0.150 $[-0.185, -0.120]$	<0.001
residual − residual, perm. (PCC)	20%	-0.001 $[-0.003, +0.002]$	1.000
residual − per-slide best (PCC)	20%	-0.005 $[-0.007, -0.003]$	0.003
value-cond. − value-cond., constant (recall)	20%	$+0.074$ $[+0.032, +0.118]$	0.038
value-cond. − feasible (PCC)	20%	-0.163 $[-0.215, -0.123]$	<0.001

E Spiked positive control

Table 11: Semi-synthetic control on the 22 slides: residual policy (and its permuted-feedback version) minus the feasible rule, by spike amplitude and budget (PCC; unadjusted 95% CIs resampling studies). The registered success criterion was a positive margin with a CI excluding zero at $\alpha = 2$.

α	Budget	spiked genes in top 50	residual − feasible	residual, perm. − feasible	residual − residual, perm.
0	5%	0	-0.015 $[-0.023, -0.008]$	-0.026 $[-0.034, -0.018]$	$+0.010$ $[+0.002, +0.019]$
0	10%	0	-0.007 $[-0.011, -0.003]$	-0.008 $[-0.012, -0.003]$	$+0.001$ $[-0.002, +0.004]$
1	5%	20	-0.022 $[-0.031, -0.013]$	-0.030 $[-0.038, -0.023]$	$+0.007$ $[-0.001, +0.015]$
1	10%	20	-0.010 $[-0.017, -0.004]$	-0.013 $[-0.019, -0.007]$	$+0.003$ $[-0.002, +0.007]$
2	5%	20	-0.021 $[-0.030, -0.012]$	-0.034 $[-0.043, -0.028]$	$+0.013$ $[+0.004, +0.023]$
2	10%	20	-0.011 $[-0.019, -0.004]$	-0.018 $[-0.026, -0.010]$	$+0.007$ $[-0.000, +0.014]$

F Platform and tissue splits

The platform split is exploratory and the per-tissue means are descriptive; both use stage-1 margins.

Table 12: Selected stage-1 margins by platform (unadjusted study-level sign-flip p) and by tissue (mean).

	HD (n=7, 5 cl.)	Visium (n=15, 13 cl.)	prostate (n=6)	kidney (n=4)	lung (n=6)	colon (n=6)
Recon., error-red. (1%)	-0.113 (0.062)	-0.068 (<0.001)	-0.078	-0.085	-0.081	-0.088
Recon., value-cond. (10%)	-0.103 (0.062)	-0.174 (<0.001)	-0.115	-0.171	-0.141	-0.185
Detect., value-cond. (10%)	$+0.048$ (0.062)	$+0.008$ (0.426)	-0.001	$+0.022$	$+0.036$	$+0.027$
Detect., value-cond. (20%)	$+0.099$ (0.062)	$+0.015$ (0.469)	$+0.015$	$+0.024$	$+0.052$	$+0.070$
Detect., rounds of 5 (10%)	$+0.049$ (0.250)	$+0.026$ (0.044)	$+0.016$	$+0.012$	$+0.046$	$+0.053$

G Gene-set sensitivity

Table 13: Stage-1 local-variance minus best coverage reconstruction margin (PCC) under three gene sets, 22 slides, unadjusted 95% CIs resampling studies (secondary).

Budget	per-slide top-50 HVG	reference-subset top-50 HVG	all filtered genes
2%	-0.136 $[-0.171, -0.106]$	-0.134 $[-0.168, -0.103]$	-0.032 $[-0.047, -0.021]$
5%	-0.148 $[-0.180, -0.119]$	-0.148 $[-0.182, -0.118]$	-0.052 $[-0.068, -0.040]$
10%	-0.152 $[-0.186, -0.122]$	-0.153 $[-0.189, -0.122]$	-0.068 $[-0.087, -0.052]$
20%	-0.165 $[-0.217, -0.125]$	-0.166 $[-0.217, -0.126]$	-0.079 $[-0.101, -0.061]$

H Imputer comparison on the earlier slide set

The k-NN and kriging imputers were run only on the earlier slide set. Table 14 reports error-reduction minus the per-slide best non-adaptive rule on the five human slides from that run, whose regions held one pseudo-spot each, with the number of slides on which the margin was positive. With five slides these values are descriptive.

Table 14: Error-reduction minus per-slide best non-adaptive rule (PCC), mean over 5 human Visium HD slides (positive/total).

Budget	kNN	kriging	graphflow
0.5%	-0.088 (0/5)	-0.088 (2/5)	-0.082 (0/5)
1%	-0.114 (0/5)	-0.138 (1/5)	-0.083 (0/5)
2%	-0.091 (0/5)	-0.150 (1/5)	-0.040 (0/5)
5%	-0.121 (0/5)	-0.141 (1/5)	-0.011 (1/5)
10%	-0.134 (0/5)	-0.133 (0/5)	-0.000 (2/5)
20%	-0.145 (0/5)	-0.126 (0/5)	-0.002 (2/5)

I Native SOFisher run

We ran SOFisher from the authors' repository on its bundled mouse cortex data for 5 million steps (seed 222) with the repository's cortex script. The only change was checkpoint and resume support, so the run could continue after interruptions. Figure 3 shows the evaluation reward, which counts captured target regions. Its median was -238 over the first 5% of training and $+147$ over the last 10% (best 514). Rare evaluation episodes have rewards near -1.7×10^5, so the figure shows rolling medians.

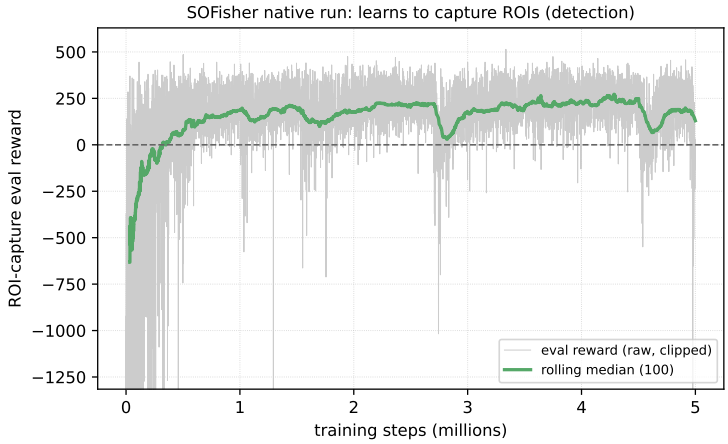


Figure 3: SOFisher evaluation reward during training (rolling median), native run.

J Configuration and implementation checks

All values are fixed in the released `config.py`. Seeds: global 2021; random policy $\{0, \ldots, 4\}$; the permutation for feedback ablation uses the global seed. Graphflow: image PCA 64, random Fourier features 512 (median-heuristic bandwidth), ridge 10, graph $k = 8$, 20 steps, step size 0.5. k-NN imputer: $k = 10$, joint image and space features. Kriging: image PCA 64, ridge 10, residual $k = 10$. Placement: feature PCA 64, median-heuristic lengthscale, noise 10^{-2}. Error-reduction and residual: neighbourhood 15; the residual policy uses the 256 genes with the highest variance among profiled regions and 10 profiled neighbours. Local-variance: spatial kernel width 0.1 of the slide diagonal, exploration weight 0.5. Spiked control: 20 genes, 3 hotspots, radius 0.05 of the diagonal, noise 0.1 of the reference standard deviation. Statistics: 5,000 bootstrap resamples; exact sign-flip enumeration over study clusters; MDE from 300 study resamples with 3,000 sign flips each.

The code for the larger slides replaces dense distance computations with sparse and incremental ones; a 132-run check (11 policies, 3 imputers, 2 slide sizes, 2 budgets) reproduced the original code's selections and scores exactly. In the stage-2 run, the five non-adaptive rules reproduced the stage-1 per-slide scores exactly, and the local-variance and diversity policies reproduced their stage-1 means. A caching error interrupted the first stage-2 run. For the results saved before it, a stale cache entry could only have produced a crash, and the error was fixed before any result was analysed.

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